

Causal Inference: Foundations & Applications

Chapter 2: Randomized Experiments

Applied Use Case: Did the Retention Offer Actually Reduce Churn?

1. Why Randomized Experiments (A/B Tests) Work

In observational CRM data, retention offers are assigned non-randomly based on customer characteristics, creating confounding bias. A **randomized experiment (A/B Test)** breaks the dependency between customer attributes and treatment assignment.

Randomization ensures that missing counterfactual outcomes are **Missing Completely at Random (MCAR)**, thereby making the treated ($A = 1$) and untreated ($A = 0$) groups structurally comparable.

2. Marginal Exchangeability ($Y^a \perp\!\!\!\perp A$)

When retention offers are assigned via a fair random mechanism (e.g., coin flip), the potential outcomes ($Y^{a=1}, Y^{a=0}$) become independent of actual treatment assignment A :

$$Y^a \perp\!\!\!\perp A \quad \text{for } a \in \{0, 1\}$$

Consequence: Association Equals Causation

Under marginal exchangeability, the observed churn rate among customers receiving the offer equals the counterfactual churn rate if the entire population had received the offer:

$$\Pr[Y = 1 \mid A = 1] = \Pr[Y^{a=1} = 1]$$

$$\Pr[Y = 1 \mid A = 0] = \Pr[Y^{a=0} = 1]$$

Therefore, the observational risk ratio equals the causal risk ratio:

$$\frac{\Pr[Y = 1 \mid A = 1]}{\Pr[Y = 1 \mid A = 0]} = \frac{\Pr[Y^{a=1} = 1]}{\Pr[Y^{a=0} = 1]}$$

3. Conditional Randomization & Conditional Exchangeability

In business settings, companies rarely randomize treatment uniformly across all customers. Instead, higher-risk customer segments receive offers with a higher probability.

Experimental Design Types

1. **Marginally Randomized Experiment:** Every customer has the exact same probability of receiving an offer (e.g., 50% for all). Yields **marginal exchangeability**.
2. **Conditionally Randomized Experiment:** Treatment probabilities vary by customer segment L (e.g., 75% offer probability for high-risk customers $L = 1$, and 50% for low-risk customers $L = 0$).

Conditional Exchangeability ($Y^a \perp\!\!\!\perp A \mid L$)

While treated ($A = 1$) and untreated ($A = 0$) groups are not exchangeable across the entire population, they are **exchangeable within each stratum of covariate L** :

$$Y^a \perp\!\!\!\perp A \mid L \quad \text{for } a \in \{0, 1\}$$

4. Adjustment Method 1: Standardization (Parametric / Non-parametric g-Formula)

To recover the overall marginal causal risk $\Pr[Y^a = 1]$ from a conditionally randomized experiment, we compute a weighted average of stratum-specific risks, weighted by the proportion of each stratum in the full population:

$$\Pr[Y^a = 1] = \sum_l \Pr[Y = 1 \mid A = a, L = l] \cdot \Pr[L = l]$$

Application in Churn Analysis

1. Estimate observed churn rates within each stratum L for offered ($A = 1$) and non-offered ($A = 0$) customers.
2. Multiply each stratum-specific rate by the proportion of customers in that stratum across the entire database ($\Pr[L = l]$).
3. Sum the weighted rates to compute the standardized population counterfactual risk.

5. Adjustment Method 2: Inverse Probability Weighting (IPW) & Pseudo-Populations

Inverse Probability Weighting creates a hypothetical **pseudo-population** where the association between confounder L and treatment A is completely severed, restoring unconditional exchangeability.

Definition of Inverse Probability Weights (W^A)

Each individual receives a weight equal to the inverse of the probability of receiving the treatment they actually received:

$$W^A = \frac{1}{f(A \mid L)}$$

Step-by-Step Numerical Example (Pseudo-Population Construction)

Original Database (100 Customers)

- **High Risk** ($L = 1$): 60 customers.
 - 75% offered treatment $\implies$ 45 treated ($A = 1$), 15 untreated ($A = 0$).
- **Low Risk** ($L = 0$): 40 customers.
 - 50% offered treatment $\implies$ 20 treated ($A = 1$), 20 untreated ($A = 0$).

Calculating IP Weights ($W^A = 1/f(A | L)$)

1. High Risk ($L = 1$), Offered ($A = 1$):

$$\Pr[A = 1 | L = 1] = 0.75 \implies \text{Weight} = \frac{1}{0.75} = 1.33 \implies 45 \times 1.33 = \mathbf{60} \text{ weighted subjects}$$

2. High Risk ($L = 1$), Not Offered ($A = 0$):

$$\Pr[A = 0 | L = 1] = 0.25 \implies \text{Weight} = \frac{1}{0.25} = 4.00 \implies 15 \times 4.00 = \mathbf{60} \text{ weighted subjects}$$

3. Low Risk ($L = 0$), Offered ($A = 1$):

$$\Pr[A = 1 | L = 0] = 0.50 \implies \text{Weight} = \frac{1}{0.50} = 2.00 \implies 20 \times 2.00 = \mathbf{40} \text{ weighted subjects}$$

4. Low Risk ($L = 0$), Not Offered ($A = 0$):

$$\Pr[A = 0 | L = 0] = 0.50 \implies \text{Weight} = \frac{1}{0.50} = 2.00 \implies 20 \times 2.00 = \mathbf{40} \text{ weighted subjects}$$

Resulting Pseudo-Population (200 Weighted Subjects)

- **Treated Group** ($A = 1$): 60 High Risk (60%) + 40 Low Risk (40%) = 100 subjects.
- **Untreated Group** ($A = 0$): 60 High Risk (60%) + 40 Low Risk (40%) = 100 subjects.

Core Insight of IPW

In the pseudo-population, both $A = 1$ and $A = 0$ arms have **identical baseline risk distributions (60% High Risk, 40% Low Risk)**. Confounding by L is eliminated ($A \perp\!\!\!\perp L$ in the pseudo-population), allowing direct contrast of weighted churn rates.

6. Equivalence of Standardization and IPW

Under the **Positivity Condition** ($\Pr[A = a | L = l] > 0$ for all a, l), Standardization and IPW are mathematically equivalent:

$$\sum_l \Pr[Y = 1 | A = a, L = l] \Pr[L = l] = E \left[\frac{I(A = a) Y}{f(A | L)} \right]$$

Both methods allow data scientists and decision-makers to evaluate counterfactual scenarios (“What if everyone received the offer?” vs. “What if no one received the offer?”) using observational or conditionally randomized data.